higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO INFORMATION SYSTEMS
NQF LEVEL 2

13 NOVEMBER 2019

This marking guideline consists of 9 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	B – together		
	1.1.2	C – management		
	1.1.3	A – system maintenance		
	1.1.4	D – decision making		
	1.1.5	D – intellectual		
	1.1.6	A – abacus		
	1.1.7	B – delete		
	1.1.8	C – scanner		
	1.1.9	D – num lock		
	1.1.10	C – cutter plotter		
			(10 × 1)	(10)
1.2	1.2.1	Organising		
	1.2.2	Strategies		
	1.2.3	Procedures		
	1.2.4	Free		
	1.2.5	Antivirus		
	1.2.6	Electricity		
	1.2.7	Binary		
	1.2.8	Input		
	1.2.9	Technology		
	1.2.10	Arrow		
			(10 × 1)	(10)
1.3	1.3.1	True		
	1.3.2	True		
	1.3.3	False		
	1.3.4	False		
	1.3.5	True		
	1.3.6	False		
	1.3.7	False		
	1.3.8	True		
	1.3.9	True		
	1.3.10	False		
			(10 × 1)	(10)
1.4	1.4.1	H – critical thinking		
	1.4.2	A – information system		
	1.4.3	J – knowledge base		
	1.4.4	B – risk		
	1.4.5	D – firewall		
	1.4.6	F – SRAM		
	1.4.7	G – SD card		
	1.4.8	I – resolution		
	1.4.9	E – arithmetic logic unit		
	1.4.10	C – system files		
			(10 × 1)	(10)

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1.5 1.5.1 Decision support system
1.5.2 General manager
1.5.3 Frequently asked questions
1.5.4 Uninterrupted power supply
1.5.5 Management information system
1.5.6 Read-only memory
1.5.7 Personal computer
1.5.8 Central processing unit
1.5.9 Dynamic random access memory
1.5.10 Kilobyte

(10 × 1) (10)
[50]

SECTION B

QUESTION 2

2.1
- A group of people
- Working together in an organised way
- To achieve one or more goals (Any 2 × 1) (2)

2.2
- Finance/accounting
- Administration
- Human resources
- Information technology
- Marketing/advertising
- Sales (Any 2 × 2) (4)

2.3
- Planning helps decide in advance the course of action to achieve goals.
- Planning allows the manager to plan for possible setbacks and come up with emergency plans.
- The manager can allocate tasks to suitable people ahead of time.
- The plan keeps the organisation's goals in mind. (Any 2 × 2) (4)

2.4
- Reorder stock
- Coordinate staff work schedules
- Monitor staff
- Settle complaints/grievances
- Deal with customers (Any 3 × 2) (6)

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2.5
1. Define the problem
   - The problem should be accurately identified.
2. Gather information
   - Obtain more information relating to the problem.
3. Brainstorm solutions
   - Develop possible solutions with a team.

ALTERNATIVE FOR STEP 3:

3. Analyse the information
   - Look closely at the information that has been gathered, taking all the facts into account.

STEPS IN CORRECT ORDER (1 × 1)
FIRST THREE STEPS (3 × 1)
DESCRIPTION OF EACH STEP (3 × 1) (6)

[22]

QUESTION 3

3.1
- Data are the raw facts or figures.
- Once the data are processed or manipulated and given context, it becomes information. (4)

3.2
- A knowledge base can be used to facilitate training of new employees by creating a database of information to provide ongoing support.
- A knowledge base can be used to improve customer experience by providing helpful information or answering frequently asked questions as customer support.
- A knowledge base can facilitate existing employees by providing access to policies or other documents.
- A knowledge base can be used to track customers' habits and in turn offer them relevant products or special offers. (Any 2 × 2) (4)

3.3
- Information systems are able to collect and store data about anything on which it is designed to collect data.
- Data within the information system can be used to get a competitive advantage in the market.
- An information system is versatile and can be used to provide management with reports that assist in making decisions. (4)

3.4 3.4.1
- Data
- Procedures
- People
- Networks
- Hardware
- Software (Any 5 × 2) (10)

3.4.2
- The system must be designed in a modular way.
- The cost needs to be considered and negotiated if necessary.
- The system must be reliable.
- The system must be able perform adequately.
- The system needs to be flexible.
- The system should be easy to maintain. (Any 1 × 1) (1)

[23]

QUESTION 4

4.1 4.1.1 A centralised information system.

4.1.2 A decentralised information system.

(2 × 2) (4)

4.2 Financial MIS will be used to prepare reports on data that has been analysed.

(2)

4.3
- Mandatory access control
  - A type of access control implemented by the system.
  - Access is granted based on the type of security clearance given to the person accessing the information.

  (Any 1 × 2)
- Discretionary access control.
  - This type of access control is implemented by the file owner.
  - The attributes specify the type of access other users have to the file.

  (Any 1 × 2) (4)

4.4
- Intentional threats
- Unintentional threats (2 × 2) (4)

4.5
- This type of attack takes the system hostage.
- An attacker will block the computer system or its files through encryption.
- The attacker then demands a ransom in order to unblock the system. (6)

4.6
- It can be used to aid teaching in the lecture venue.
- It can be used for research.
- It can be used for self-paced learning.
- It can be used with applications to notify students of updates on campus.
- It can be used for e-learning. (Any 3 × 2) (6)

4.7
- Trojans
- Worms
- Logic bombs
- Ransomware (Any 2 × 2) (4)

[30]

QUESTION 5

5.1
- They can take pictures.
- They can be used to explore applications.
- They can be used for productivity.
- They can be used to watch videos.
- They can play music.
- They can be used to surf the internet. (Any 3 × 2) (6)

5.2 Artificial intelligence is concerned with developing machines that are able to✓✓ think like humans beings and make logical decisions.✓✓ (4)

5.3
- Smartphone
- Tablet
- Smart TV
- Smart watch
- Smart fridge (Any 2 × 2) (4)

5.4 5.4.1
- Input peripheral devices are used to enter data into the computer.
- Output peripheral devices are used to provide feedback to the user. (4)

5.4.2 Track pad (1)

5.5 5.5.1 Time sharing mode (2)

5.5.2
- Time sharing mode allows multiple users to process tasks simultaneously.
- Processing time is divided among users so that everyone gets the same amount of usage time. (Any 1 × 2) (2)

[23]

QUESTION 6

6.1
- A stored program is a program stored on the computer's hard drive.
- It is able to take input from users.
- It then runs instructions on the computer. (6)

6.2
- Computers are quicker to process information.
- Information is stored in once place.
- Computers allow people to connect with one another.
- Computers are useful educational tools.
- Computers are reliable and robust.
- When used with the internet, computers allow files to be shared.

(Any 3 × 2) (6)

6.3 The purpose of the system unit is to hold all the other components in place and to protect them. (2)

6.4 The F1 key is used for Windows help. (2)

6.5
- Primary storage is volatile whereas secondary storage is more stable.
- Primary storage is used as working memory and secondary storage is used for long-term storage.
- Primary storage is more expensive than secondary storage per megabyte.
- Primary storage is faster to access than secondary storage. (Any 2 × 2) (4)

6.6 6.6.1 A multifunction printer can be used instead of two separate machines. (2)

6.6.2
- Multifunction printers can be costly.
- If one component fails, the entire machine needs to go in for repairs.
- The machine will queue jobs on a first-come-first-serve basis.

(Any 2 × 2) (4)

6.7 6.7.1 The files can be compressed. (Any 1 × 2)

6.7.2
- Files are encoded more efficiently
- File size is therefore reduced (Any 1 × 2)

(2 × 2) (4)

6.8

CD-RW			DVD-R		
1.	Store files	✓✓	1.	Store movies	✓✓
2.	Store music	✓✓	2.	Back-up system	✓✓

(8)

6.9 6.9.1 Keyboard entry would be used when using a personal computer or laptop.

6.9.2 Direct entry would be used when a scanner is available to scan a bar code.

(2 × 2) (4)

6.10 The operator at an airline reservation desk✓ captures the clients' information, but the processing will be done at a central server.✓✓ (3)

6.11 6.11.1
- A hard drive is fairly cheap and reliable.
- They have large storage capacity.
- Their cost per megabyte is cheap when compared to other storage devices. (Any 2 × 2) (4)

6.11.2 If the disk fails there is no backup available. (1)

6.11.3
- RAID level 1
- Mirroring (Any 1 × 2) (2)

[52]

TOTAL SECTION B: 150
GRAND TOTAL: 200

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